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Data Engineering and Visualization, Winter Semester 2026
Dataset Catalog For the project

Dataset Catalog: The catalog has recommended datasets from both domains, together with their potential integration fields. Use it as a starting point to identify datasets that may share meaningful geographic or temporal attributes before making your final selection. The catalog is not an exclusive list; you are encouraged to explore other datasets within the two domains, provided they meet the project requirements.

Housing & Development

ID	Dataset	Key Integration Fields
H1	DOB Safety Violations	Borough, Latitude/Longitude, Violation Date, Inspection Date
H2	Affordable Housing Production by Building	Borough, Street Name, Address, ZIP Code, Completion Date
H3	DOB NOW: Build Approved Permits	Borough Name, Street Name, ZIP Code, Permit/Approval Dates
H4	DOB Job Application Filings	Borough, Block, ZIP Code, Latitude/Longitude, Application/Action Dates
H5	DOB NOW: Build Job Application Filings	Borough, Block, Lot, ZIP Code, Latitude/Longitude, Filing Dates
H6	Housing Maintenance Code Complaints and Problems	Borough, ZIP Code, Address/Location, Complaint Date

Transportation

ID	Dataset	Key Integration Fields
T1	Parking Meters Locations and Status	Borough, Latitude/Longitude, Meter Location/Street
T2	Traffic Volume Counts Historical	Street Name, Borough, Count Location, Count Date
T3	Street Construction Permits (2022–Present)	Borough Name, Street Name, Permit Dates, Location
T4	SIRTA Data	Crash Borough, ZIP Code, Latitude/Longitude, Crash Date
T5	Bus Breakdown and Delays	Boro, Location, Occurred_On
T6	Yellow Taxi Trip Data	Pickup/Drop-off Location, Pickup/Drop-off Date/Time, Distance
T7	Green Taxi Trip Data	Pickup/Drop-off Location, Pickup/Drop-off Date/Time
T8	NYCDCP Manhattan Bike Counts	Latitude/Longitude, Start/End Station, Start/End Date/Time

Note: The listed fields indicate potential integration attributes, not necessarily direct join keys. The selected datasets do **not need to have the same number of records or a one-to-one relationship**. Datasets may be integrated using shared geographic attributes (e.g., Borough, ZIP Code, street name,or

coordinates), common time periods, or spatial proximity. When direct matching is not possible, teams may aggregate or transform one or both datasets to a common geographic or temporal level. The chosen granularity must be appropriate for the research question and clearly justified. Teams should not perform a row-level join merely because a common attribute exists. Teams must clearly explain how the selected datasets are integrated and how the integration supports their research questions.